

Autumn (Odd) Semester Examination -2025

Course Code: PHYUGMCC2305

Course Title: Thermal Physics

Department of Appearing Students: PHYSICS, SEM-III, 2nd Year

Full Marks: 60

Time: 2:30 hrs.

[Use separate sheets for answering Group-A and Group-B questions.]

GROUP- A

(Answer any FIVE questions)

1. (a) The velocity distribution function of a group of N particles is given by

$$dN_v = kv^2 dv \quad \text{for } 0 \leq v \leq V$$

$$= 0 \quad \text{for } v > V$$

Calculate the constant k in terms of N and V , and also calculate the mean velocity and RMS velocity of the particle. 2+2+2

2. (a) Write down the van der Waals equation of state for a real gas and explain the significance of the correction terms. (b) Find the critical constants (V_c , P_c , T_c) of a real gas obeying the van der Waals equation of state? (1+2) +3

3. (a) Show that with the increase of volume per mole, the van der Waals equation of state tends to that of an ideal gas. (b) What is the Boyle temperature? How is it related to critical temperature? 2+(1+3)

4. (a) A rigid triangular molecule consists of three non-collinear atoms joined by rigid rods. Using the laws of equipartition of energy, obtain the average energy and the constant-pressure molar specific heat C_p of an ideal gas consisting of such molecules.

(b) In 1- dimension, an ensemble of N classical particles has energy of the form, $E = \frac{1}{2m}(p_x^2) + \frac{1}{2}kx^2$ where m is the mass of the particle, k is a constant. Find the average internal energy of the system at temperature T . 3+3

5. (a) Write down the expression (no derivation) for the coefficient of viscosity and thermal conductivity of a gas and find the relation between them. (b) Explain how the coefficient of viscosity (η) varies with Pressure and temperature? Explain. 2+ (2+2)

6. Two boxes, A and B, contain an equal number of molecules of the same gas. If the volumes are V_A and V_B , find the ratio of their respective mean free paths. (b) A narrow molecular beam enters a vessel filled with gas under low Pressure. Find the mean free path of the molecules if the beam intensity decreased to $1/n$ times its initial intensity over a distance Δl . 3+3

3x3=2

36

GROUP- B

(Answer Question No. 7 and any three from the remaining)

7. a) An oil bath kept at 27°C is being supplied heat at the rate of 100Js^{-1} . Assuming the process to be quasi-static, calculate the rate of increase of entropy of the system. 4
- b) With the use of a suitable diagram prove that working between same two temperatures no other engine can be more efficient than a reversible Carnot engine. 3
- c) An engine absorbs heat at temperature of 1000K and rejects heat at 600K . Assuming the engine operating at maximum possible efficiency, estimate the amount of work performed by the engine for 200J heat input. 3
- d) Check whether the relations $\left(\frac{\partial H}{\partial S}\right)_P = T$ and $\left(\frac{\partial H}{\partial P}\right)_S = V$ hold true or not for the thermodynamical potential called enthalpy, $H = U + PV$. 2
- e) From the first law of thermodynamics show that for an ideal gas $C_P - C_V = R$, where symbols have their usual meanings. 3
8. With a suitable diagram of Carnot cycle, derive an expression for the efficiency of car reversible Carnot engine explaining all the processes involved in the cycle. Can the efficiency of a Carnot engine be 100%? Justify your response. 4+1
9. When a mass m of water at temperature $T_1\text{K}$ is isobarically and adiabatically mixed with an equal amount of water at $T_2\text{K}$, then show that the entropy change of the universe is $2mC_p \log \left(\frac{T_1+T_2}{2\sqrt{T_1T_2}}\right)$. 5
10. With a suitable diagram explain the thermodynamic processes involved in the diesel cycle. How a diesel cycle differs from an Otto cycle? 4+1
11. Derive Maxwell's thermodynamical relations: $\left(\frac{\partial P}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$ and $\left(\frac{\partial V}{\partial T}\right)_P = -\left(\frac{\partial S}{\partial P}\right)_T$. You may use the expressions of thermodynamical potentials: $F=U-TS$ and $G=H-TS$ respectively. Symbols have their usual meanings. 2+3
12. a) What are extensive in intensive variables? (b) What do you mean by a thermodynamic scale of temperature? Is it possible to attain absolute zero on the thermodynamic scale of temperature? Justify your answer. 2+(2+1)

Autumn Semester Examination - 2025

Course Code: PHYUGSEC2303

Course Title: Data analysis and Scientific writing

Department of Appearing Students: Physics, Semester -III, 2nd Year

Full Marks: 60

Time: 2:30 hrs

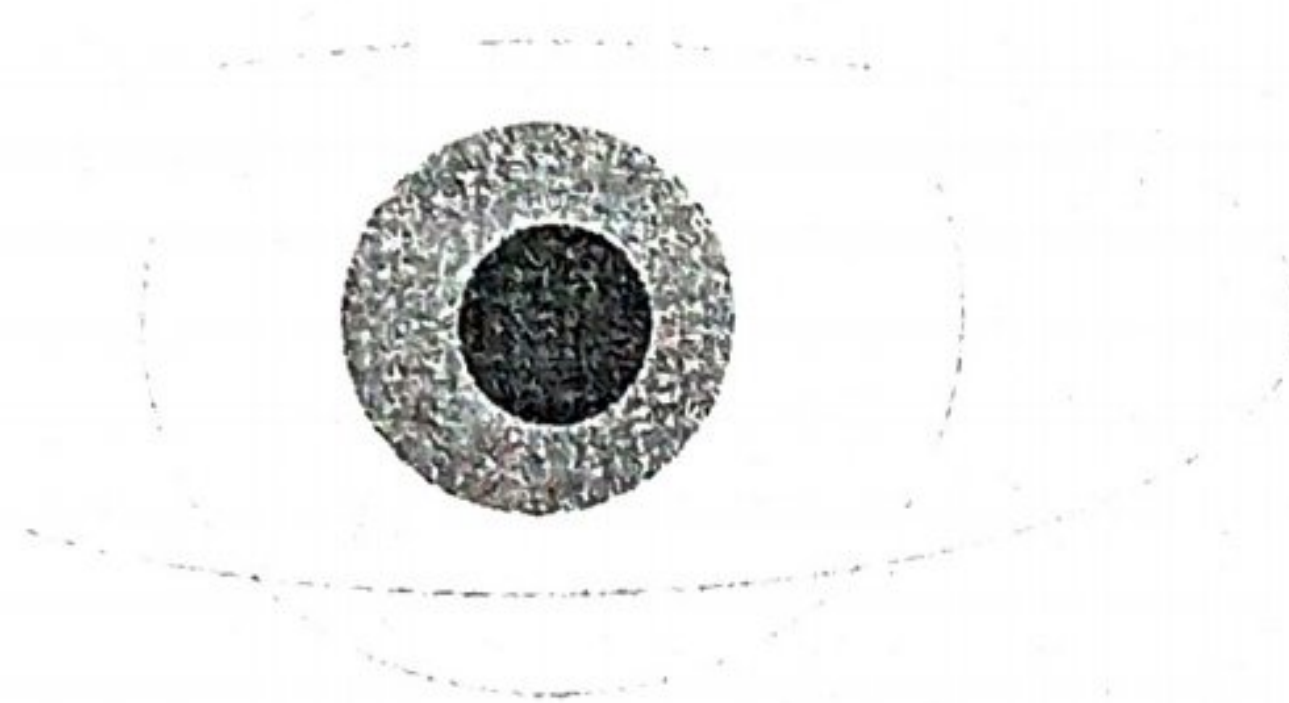
[Read the instructions carefully before answering and attempt questions as directed]

1. Write down the LaTeX commands to generate the following Table: [10]

Table 1. Health data of three individuals

Name	Age (yr)	Height (m)	Weight (kg)
Sachin Tendulkar	54	1.62	68
Baichung Bhutia	55	1.54	62
Diego Maradona	76	1.66	79
Zohran Mamdani	45	1.72	76

2. What are the steps to be followed to insert the following shape in a LibreOffice document? [10]



3. Which application under LibreOffice is suitable for calculation of the average age, average height, and average weight of the four persons whose data are presented in Table 1 (See Question No. 1)?

Describe the steps to be followed to accomplish the computation. [2+8=10]

4. Write down the LaTeX source code to produce an article having following components:

Article Title, Author Name, Author Address, Introduction, Method, Results & Discussions, Conclusion, References. [20]

5. Write the LaTeX command to generate the following: [3+7=10]

(a) Numbered equation: $x^3 + y^{3/2}z = \Sigma$ (1)

(b) Numbered array of three equations:

$$\begin{array}{lcl} x^3 + y^{3/2}z & = & \Omega \\ rx^{3/2} + qy^{3/2} & = & \omega \\ px^3 + my^3z & = & \eta \end{array} \quad (2)$$

omega

ALIAH UNIVERSITY
Autumn Semester End-term Examination-2025
Subject-Ordinary Differential Equations
Subject Code: MATUGMCC2305/ MATUGMIN2305, Full Marks-80, Duration- 3 Hrs.

Unless stated all notations have their usual meanings.

1. Answer any ten questions from the following: $10 \times 2 = 20$

- (a) Check whether the functions $x, \cos x, \cos 2x$ are linearly independent? 2
- (b) Define an oblique trajectory. 2
- (c) Find the differential equation of all circles which pass through the origin and whose centres are on the y -axis. 2
- (d) Write Lotka-Volterra model with description of the constants. 2
- (e) Define equilibrium point of a system with example. 2
- (f) Find the singular solutions of the equation $9p^2(2-y)^2 = 4(3-y)$, where $p = \frac{dy}{dx}$. 2
- (g) Solve $y = px + \log p^3$. 2
- (h) Check whether the equation $(2y^4e^y + 2xy^3 + y)dx + (x^2y^4e^y - x^2y^2 - 3x)dy = 0$ is exact. 2
- (i) Define degree and order of a differential equation. 2
- (j) Define integrating factor of a differential equation with examples. 2
- (k) Define phase portrait of a system of differential equation. 2

2. Answer any twelve questions from the following: $12 \times 5 = 60$

- (a) Find the orthogonal trajectories of the following family of curves:

i. $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$.
ii. $r^n = a^n \cos n\theta$, where a is parameter.

- (b) Solve the following differential equations:

i. $(x^2 - 1)p^2 - 2xyp + y^2 = m^2$,
ii. $p^2 - p(e^2 + e^{-2}) + 1 = 0$.

- (c) Solve $(D^2 + 1)y = \cos x \sin 3x$.

- (d) Investigate the behaviour of the system

$$\dot{x} = y, \quad \dot{y} = (\alpha - 1)x - \alpha y$$

for different values of the parameter $\alpha \in \mathbb{R}$.

- (e) Solve $(D^2 + 2)y = x^2 e^{3x}$.
- (f) Solve $(D^2 + 2D + 2)y = x^2 + \sin x$, using the method of undetermined coefficients.
- (g) Find the general solution and draw the phase portrait for the following linear system:

$$\dot{x} = -x + y, \quad \dot{y} = -y.$$

- (h) Solve $(x^3 D^3 + 2x^2 D^2 - xD + 1)y = \frac{1}{x}$.
- (i) Find a particular solution of $(x^2 + x)y'' + (2 - x^2)y' - (2 + x)y = x(x + 1)^2$ if $y = e^x$ is a solution of the corresponding homogeneous equation.

- (j) Apply the method of variation of parameters to solve the equation $(D^2 + a^2)y = \sin ax$.
- (k) Test for exactness and solve $x^2(1+x)D^2y + 2x(2+3x)Dy + 2(1+3x)y = 0$.
- (l) Transform the equation $D^2y - 2 \tan x Dy + 5y = e^x \sec x$ into normal form and hence solve.
- (m) Find all solutions of the equation

$$\dot{X} = \begin{pmatrix} 1 & -1 & 4 \\ 3 & 2 & -1 \\ 2 & 1 & -1 \end{pmatrix} X, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

- (n) Solve $(D^2 - 4D + 4)y = e^x + \cos 2x$.
- (o) Solve $(D^2 + 6D + 9)y = xe^{-3x} \sin x$.
- (p) Find the eigenvalue and corresponding eigenfunction of the following Sturm-Liouville's problem

$$\frac{d^2x}{dt^2} + \lambda x = 0, \quad x(0) = 0, \quad x'(l) = 0.$$

- (q) Solve the differential equation $(x - y + 3)dx = (2x - 2y + 5)dy$.

Autumn End Semester exam, 2025
ENGUGAEC2302 : English Communication

Full Marks: 80

Time: 3 Hours

Section I

Answer ANY TEN of the following questions:

2x10=20

1. What is *lingua franca*?
2. What is semantic gap?
3. What is passive listening?
4. What is skimming in reading?
5. What is modal verb? Give an example.
6. Define Grapevine communication.
7. What is active listening?
8. How many consonant sounds are there in English language?
9. What is homophone? Give an example.
10. What is phrase? Give an example.
11. What is intrapersonal communication?
12. What is a monologue?
13. What are diphthongs? Give an example.
14. What is LSRW ?

Section II

2. Answer **ANY FOUR** of the following questions: (word limit 150)

5x4=20

1. Distinguish between formal and informal communication. ✓
2. What is Group Discussion? Briefly explain its purpose, steps, and benefits.
3. Define body language. Elucidate some of the gestures we use to communicate. ✓
4. Write a short dialogue between two friends discussing the price hike.
5. What is the difference between summary and paraphrase? ✓
6. What are the features of close reading?

Section III

3. Answer **ANY FOUR** of the following questions. (word limit 400)

10X4=40

1. Identify the notable barriers to Effective Communication. Discuss the means to overcome those barriers. (5+5)
2. Write a job application for the post of a teacher in a college. Attach a Resume (5+5)
3. Define and discuss the differences between Verbal Communication and Non-verbal Communication. (10)
4. Explain the different flows of communication in business organisations. (10) ✓
5. Define Intra-personal Communication and Inter-personal Communication. Discuss the differences between the two. (5+5)
6. Write an email to request the Head of your Department for an extension of the deadline for the submission of your assignment. Cite an appropriate reason for the delay.

Autumn Semester Examination 2025

Subject: Mathematical Physics II
Full Marks: 60

Subject Code: PHYUGMCC2306
Time: 2 hrs 30 minutes

[Each question carries 10 marks, Answer any six questions]

1. Laplace's equation in two dimensions is given as,

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} = 0$$

Show that one of the possible solution can be assumed as,

$$\varphi(x, y) = (A_1 e^{kx} + A_2 e^{-kx})(B_1 \cos ky + B_2 \sin ky)$$

where A_1, A_2, B_1, B_2 and k are constants. Now consider the following boundary conditions (i) $\varphi(x, 0) = 0$, (ii) $\varphi(x, a) = 0$, (iii) $\varphi(x \rightarrow \infty, y) = 0$, and (iv) $\varphi(0, y) = F(x, y)$. Show that for the above conditions, the solution can be generalized as,

$$\varphi(x, y) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi y}{a} e^{-\frac{n\pi x}{a}} = \sum_{n=1}^{\infty} \left[\frac{2}{a} \int_0^a F(x, y) \sin \frac{n\pi y}{a} dy \right] \sin \frac{n\pi y}{a} e^{-\frac{n\pi x}{a}}$$

(4+6)

2. (a) The Beta and Gamma functions are defined as,

$$\beta(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx \quad \text{and} \quad \Gamma(n) = \int_0^{\infty} e^{-x} x^{n-1} dx$$

Prove that

(i) $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$

(ii) $\int_0^{\infty} x^n \exp(-kx^2) dx = \frac{1}{2k^{n+1}} \Gamma\left(\frac{n+1}{2}\right); \quad k > 0$

- (b) Show that a periodic function satisfying the Dirichlet's conditions, can be represented in a complex Fourier series as

$$F(x) = \sum_{n=-\infty}^{+\infty} c_n \exp(inx); \quad \text{where, } c_n = \frac{1}{2\pi} \int_{-\pi}^{+\pi} f(x) \exp(-inx) dx$$

[(2+2)+6]

3. The Airy's differential equation is given as $\frac{d^2 y}{dx^2} + n^2 xy = 0$, where n is an arbitrary integer.

- (a) Test the singularity of the differential equation.
(b) Considering the trial solution as $y = \sum_{k=0}^{\infty} a_k x^{p+k}$, obtain the indicial equation and find the possible values of p .
(c) Derive the recurrence relation between the power series coefficients (a_k 's) and find the general solution of the differential equation.
- (1+3+6)

4. (a) A function $f(x)$ is expanded in a Hermite series $f(x) = \sum_{n=0}^{\infty} a_n H_n(x)$. Show that

$$a_n = \frac{1}{2^n \sqrt{\pi n!}} \int_{-\infty}^{+\infty} f(x) H_n(x) e^{-x^2} dx$$

- (b) Establish the relation:

$$J_n(x) J'_{-n}(x) - J_{-n}(x) J'_n(x) = -\frac{2 \sin n\pi}{\pi x}$$

where, J_n is the Bessel's polynomial.

(5+5)

5. (a) Solve by the method of separation of variables,

$$3 \frac{\partial \varphi}{\partial x} + 2 \frac{\partial \varphi}{\partial y} = 0; \quad \varphi(x, 0) = 4e^{-x}$$

Handwritten notes:
 $m^2 - k^2 = 0$
 $m^2 = k^2$
 $m = \pm k$
 $\therefore (D \pm k)$

- (b) Find the Fourier series corresponding to the periodic function $f(x)$ defined in the interval $-\pi < x < +\pi$ as

$$f(x) = \begin{cases} 0 & -\pi < x \leq 0 \\ \frac{\pi x}{4} & 0 < x \leq \pi \end{cases}$$

and hence deduce that

$$\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

[3+(5+2)]

6. Consider the differential equation $\frac{d}{dx} [(1-x^2)y'] + (\alpha + \beta x^2)y = 0$; where α and β are positive entity.

- (a) Test the singularity of the differential equation.
 (b) Considering the Fuch theorem, comment on the results obtained in (a)
 (c) Considering the trial solution as $y = \sum_{k=0}^{\infty} a_k x^{p+k}$, obtain the indicial equation and find the possible values of p .
 (d) Derive the recurrence relation between the power series coefficients (a_k 's) and show that the solution for larger root of the differential equation can be expressed as,

$$y(x) = a_0 x \left[1 + \frac{2-\alpha}{6} x^2 + \left\{ \frac{(2-\alpha)(12-\alpha)}{120} - \frac{\beta}{20} \right\} x^4 + \dots \right]$$

[1+1+3+5]

7. (a) Given the generating function of the Legendre polynomial $P_n(x)$ as,

$$(1 - 2xt + t^2)^{-\frac{1}{2}} = \sum_n P_n(x) t^n$$

find $P_{25}(-1)$ and $P_{100}(-1)$. Comment on your result.

- (b) What is the condition for linear independence of two functions? Find the two linearly independent solutions of the Cauchy-Euler type differential equation,

$$r^2 \frac{d^2 R}{dr^2} + 2r \frac{dR}{dr} - l(l+1)R = 0$$

[(3+1)+(1+5)]

8. (a) Assume $+q$ and $-q$ charges are placed at $z = -a$ and $z = +a$ respectively. Prove that the expansion of the potential due to such linear electric dipole starts with $P_1(\cos \theta)$.
 (b) Given the generating function for the Laguerre polynomial $L_n(x)$ as,

$$\frac{1}{1-t} \exp\left(-\frac{xt}{1-t}\right) = \sum_{n=0}^{\infty} \frac{L_n(x)}{n!} t^n$$

Prove that (i) $L'_n(0) = -n$ (ii) $L''_n(0) = \frac{1}{2}n(n-1)$

[4+(3+3)]

$p+k/2-2$

$p+k-1$

Aliah University

Department of Economics

Autumn (Odd) Semester Examination, December 2025

For the 2nd Year 3rd semester students

Paper Name-Development Studies

Paper Code-ECOUGMDC2303

Full Marks-60

Time- $\frac{2\frac{1}{2}}$ Hours

Group-A

Answer any five

(5x2=10)

1. What do you mean by sustainable development?
2. What is the headcount measure of poverty? What are its limitations?
3. How can you find NNP_{FC} from GDP_{MP} ?
4. Differentiate between public good and private good.
5. Differentiate between real GDP and Nominal GDP.
6. What is a merit good?
7. What is the concept of Green GDP?

Group B

Answer any four

(4x5=20)

8. Differentiate between economic growth and economic development.
9. What are the negative impacts of high unemployment on an economy?
10. Explain the free rider problem and the tragedy of common?
11. Explain the vicious cycle of poverty.
12. Differentiate between:
 - a. Voluntary Unemployment and Involuntary unemployment.
 - b. Disguised Unemployment and Frictional Unemployment. (2.5 x2 = 5)

Group C

Answer any three

(3x 10 =30)

13. Briefly discuss the Sustainable development Goals of the United Nation. (10)
14. a. Explain the concept of Human Development Index.
b. Explain the concept of Multidimension Index of poverty. (5+5)
15. a. Do you think national income or GDP is a good indicator of welfare for an economy? Discuss. (5)
b. Explain the 4-sector circular flow diagram with leakages and injection. (5)
16. Explain positive and negative externalities in case of both consumption and production.
17. A. What are the causes of demand pull and cost push inflation? (5)
B. Explain the strategy used by government of India to tackle poverty and unemployment. (5)
